

Heart Disease Risk Factor Analysis

Problem Statement & Project Overview

1. Problem Statement

Cardiovascular disease is one of the leading causes of death worldwide, and early identification of at-risk individuals can significantly improve treatment outcomes. Healthcare providers routinely collect patient data such as age, cholesterol levels, blood pressure, and exercise response, but this data is often not analyzed systematically to uncover patterns that could support early diagnosis. This project aims to analyze a clinical dataset of patient health records to identify the key factors most strongly associated with the presence of heart disease, and to present these findings through an end-to-end data analytics workflow.

2. Objective

To perform an end-to-end data analysis on patient health records using Python, SQL, and Power BI in order to identify the demographic and clinical factors most strongly associated with heart disease, and to present these insights through an interactive dashboard that can support data-driven screening decisions.

3. Key Questions This Project Answers

- What is the overall prevalence of heart disease in the patient population?
- Which age group carries the highest risk of heart disease?
- Is there a significant difference in disease rate between male and female patients?
- Which chest pain type is most strongly associated with a heart disease diagnosis?
- Do patients with heart disease show higher average cholesterol levels?
- Does exercise-induced angina correlate with a higher disease rate?
- Does elevated fasting blood sugar increase the likelihood of heart disease?
- How does resting blood pressure relate to heart disease risk?
- Is maximum heart rate during exercise lower in patients with heart disease?
- How does the number of major vessels affect disease risk?

4. Dataset Description

Dataset: UCI Heart Disease Dataset (Kaggle) | **Records:** 302 unique patients (post-cleaning) | **Attributes:** 14

Column	Description
age	Patient age (years)
sex	Gender of the patient
cp	Chest pain type (4 categories)

trestbps	Resting blood pressure (mm Hg)
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar > 120 mg/dl
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
ca	Number of major vessels colored by fluoroscopy
target	Presence of heart disease (1 = Yes, 0 = No)

5. Tools & Methodology

- **Python (Pandas, Matplotlib, Seaborn)** — Data cleaning, exploratory data analysis, and visualization
- **SQL (MS SQL Server)** — Business-question-driven querying for aggregated clinical insights
- **Power BI** — Interactive dashboard for stakeholder-facing exploration of key metrics

6. Expected Outcome

A cleaned and validated dataset, a documented set of SQL-driven business insights, and an interactive Power BI dashboard that highlights the demographic and clinical factors most associated with heart disease — enabling faster, data-informed identification of high-risk patient profiles.